

Scalability & Future Plan

Date	5 July 2026
Team ID	XXXXXX
Project Name	Opti Crop – Smart Agricultural Production Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only a limited crop dataset	Crop recommendations are limited to available crops	High
2	No real-time weather integration	Predictions depend on manually entered weather values	High
3	No user login or history tracking	Users cannot save or view previous predictions	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small number of users	Deploy on a cloud platform (Render, AWS, Azure) with load balancing
2	Data Storage	Uses a local dataset	Store data in MySQL or MongoDB for better scalability
3	Performance	Single machine prediction	Optimize the ML model and use caching for faster predictions
4	Security	Basic Flask application	Add user authentication, HTTPS, and secure API access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
-------	-------------------------------	-----------------	-----------------

Phase 2	Real-time weather API integration	2–3 Months	More accurate crop recommendations
Phase 3	Fertilizer recommendation and disease detection	4–6 Months	Complete farming assistance
Phase 4	Mobile application with multilingual support	6–12 Months	Better accessibility and wider farmer adoption